Air Quality Index Prediction (AQI)

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I. ABSTRACT

Air pollution has become one of the main concerns for our environment and health. The levels of pollutants are consistently increasing in the urban areas, which makes it even more concerning. To measure air pollution, the index called Air Quality Index(AQI) is being considered. AQI helps to get insight into the impact of air pollution on human health.

In this research, machine learning models would be applied to predict AQI. The models used in this project are Linear Regression, Support Vector Machine (SVM), XGBoost, and CatBoost. In this preprocessing stage, the missing data, scaling and splitting datasets will be addressed.

The effectiveness of the presented models would be estimated with the help of RMSE and R2 score metrics. As we can see, the best result was achieved with the XGBoost model, lowest RMSE was 16.55, and the R2 score is equal to 0.9775.

This study shows that ensemble and boosting-based models are more effective for AQI prediction because they can capture complex patterns in environmental data. These findings can help in developing better air quality monitoring systems and supporting pollution control decisions.

II. INTRODUCTION

Air pollution has become major environmental issue in many countries. The Air Quality Index(AQI) is a widely used indicator to measure the level of air pollution and its potential health effects. High AQI values indicate poor air quality that may affect human health. The increasing number of vehicles, industrial emissions, construction activities, and burning of fossil fuels have significantly deteriorated air quality in many cities. Poor air quality has severe impacts on human health, causing respiratory diseases, cardiovascular problems, and other long-term health complications. Therefore, monitoring and controlling air pollution has become an important concern for governments and environmental agencies.

The Air Quality Indes(AQI) is a standardized indicator used to measure and communicate the level of air pollution in the atmosphere. AQI converts complex air pollution data into a simple numerical value that indicates how clean or polluted the air is and the associated health risks. The AQI is calculated based on the concentration of several pollutants, including particulate matter(PM2.5 and PM10), nitrogen dioxide(NO2), sulfur dioxide(SO2), carbon monoxide(CO), and ozone(O3). Higher AQI values represent poorer air quality and greater health risks.

With the advancement of machine learning techniques, predicting AQI in advance has become an active area of research. Accurate AQI prediction can help authorities take preventive measures and enable people to plan their daily activities accordingly, thereby reducing the harmful effects of air pollution.

Several machine learning algorithms have been applied for AQI prediction in recent studies. Techniques such as Random Forest, Support Vector Machine, Gradient Boosting, and Neural Networks have shown promising performance in predicting air pollution levels. These models are capable of handling high-dimensional data and capturing complex interactions among environmental agencies in monitoring air quality, issuing early warning, and implementing effective pollution control measures.

Despite the increasing number of studies on air quality prediction, there is still a need to evaluate and compare different machine learning models to improve prediction accuracy. Therefore, the objective of this study is to develop a machine learning-based model for predicting the Air Quality Index using historical environmental data. Multiple machine learning algorithms are implemented and their performance is evaluated using standard statistical metrics. The results of this study can contribute to the development of efficient air quality prediction systems and support environmental management efforts for improving air quality.

III. LITERATURE REVIEW

Air pollution is a common environmental problem in many countries. It affects human health, daily life, plants, animals,

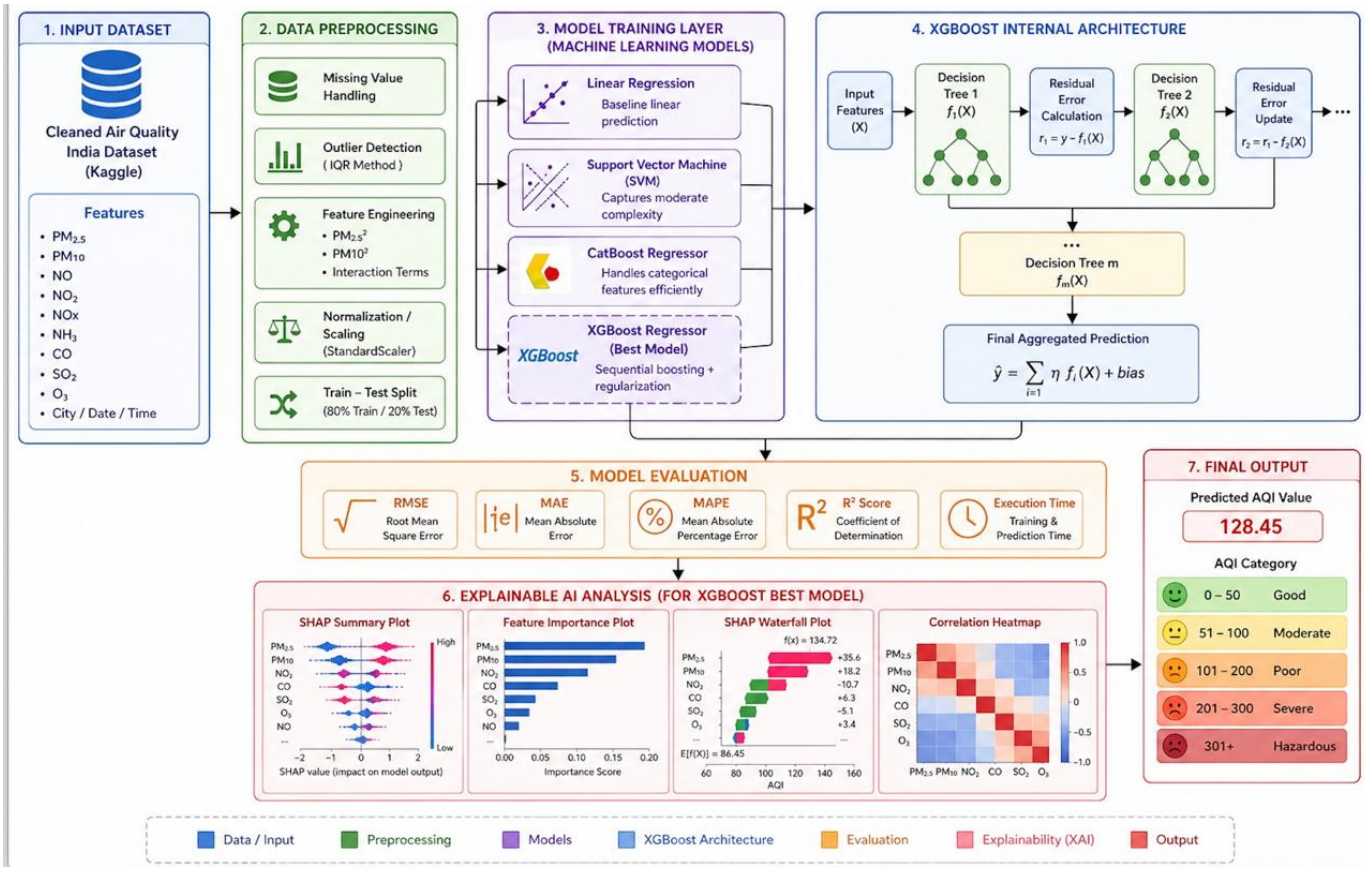


Fig. 1. Model Architecture

and climate conditions. The growth of industries, traffic, construction work, and fuel consumption has increased harmful substances in the air. Pollutants such as PM2.5, PM10, carbon monoxide, sulfur dioxide, nitrogen dioxide, and ozone are mainly responsible for poor air quality. To represent pollution levels in an easy form, the Air Quality Index (AQI) is used. AQI forecasting has become important because advance warning helps people stay safe and supports government agencies in taking preventive action. In the past, researchers used statistical and mathematical techniques to estimate air pollution levels. Those methods worked in limited conditions but often struggled when many variables were connected together. Air quality depends not only on pollutant concentration but also on weather factors such as temperature, rainfall, humidity, and wind speed. Because of these complex relationships, machine learning methods are now preferred in many studies. These techniques can study past records, learn hidden patterns, and give better future predictions. Previous health studies have explained the harmful effects of polluted air. Hajat, Hsia, and O'Neill [1] found that lower-income communities are often exposed to greater pollution levels than richer groups. Their research showed that environmental risk is unevenly distributed among society. Kelly and Fussell [2] reported that continuous exposure to polluted air may cause asthma, breathing problems, heart disease, and early death. These findings show why pollution prediction systems are necessary. Many studies have applied machine learning models for AQI forecasting. Li et al. [3] used a Long Short-Term Memory (LSTM) model for predicting pollutant concentration. Since this model can learn patterns from previous time steps, it is useful for sequential air quality data. Their results showed better performance than several traditional forecasting approaches. Random Forest is another widely used method in this area. Zhao et al. [4] applied Random Forest to estimate PM2.5 levels and achieved strong accuracy. The method combines multiple decision trees and gives a final average result. This helps reduce prediction error and improves stability. Because of these advantages, Random Forest is commonly used in environmental prediction tasks. Some researchers have focused on spatial forecasting. de Hoogh et al. [5] developed a detailed PM2.5 model using satellite information, weather data, and land-use variables. Their study showed that combining multiple sources can improve local pollution estimates. This type of work is useful for city planning and public health monitoring. Recent review papers also support the use of intelligent techniques. Samad et al. [6] examined different machine learning models for air pollution prediction and concluded that methods such as Random Forest, Support Vector Machine, XGBoost, and neural networks often perform better than standard regression models. Gryech et al. [7] explained that IoT sensors and smart

monitoring systems can be connected with machine learning for real-time AQI prediction. Hybrid deep learning methods are also gaining attention. Bai et al. [8] proposed a CNN-LSTM model for PM2.5 prediction. In this approach, CNN layers identify important features, while LSTM layers learn time-based behavior. Their combined model achieved better accuracy than using a single deep learning model.

Several additional studies have also contributed to AQI forecasting research. Singh et al. [9] applied Support Vector Machine for urban air quality prediction and found that it performs well when the dataset is properly normalized. Kumar and Goyal [10] used Artificial Neural Networks for pollutant forecasting and reported better results than traditional regression methods. Chen et al. [11] proposed an XGBoost-based prediction framework and achieved high accuracy with reduced training time. Wang et al. [12] compared multiple ensemble learning models and concluded that boosting methods generally outperform bagging methods in AQI prediction tasks. Sharma et al. [13] studied air quality prediction in Indian metropolitan cities and emphasized the importance of local climatic variables such as humidity and wind speed. Patel et al. [14] integrated IoT sensor data with machine learning models for real-time AQI monitoring and prediction, showing promising results for smart city applications.

Overall, earlier studies show that machine learning has improved AQI prediction in many regions. Models such as Random Forest, SVM, XGBoost, LSTM, ANN, and hybrid networks have given reliable results. However, model performance may change because of climate conditions, data quality, and local pollution sources. Therefore, comparing different algorithms on Indian air quality datasets is important to select the most suitable model for practical AQI forecasting.

IV. Machine Learning Models Used

We use machine learning techniques to predict Air Quality Index(AQI) based on air pollutant concentrations. The models we use are Adaboost, Linear regression, CatBoost, XGBoost, Random Forest, SVM.

A. Linear Regression

This model is a fundamental supervised learning algorithm used for predicting continuous values such as AQI. It establishes a linear relationship between independent variables (pollutant concentrations like PM2.5, NO2, CO etc.) and the dependent variable(AQI). The model assumes that changes in input features result in proportional changes in AQI values. Due to its simplicity and interpretability, Linear Regression is used as a baseline model in this study to evaluate the performance of more advanced machine learning algorithms.

$$y = b_0 + b_1x_1 + b_2x_2 + \cdots + b_ix_i \tag{1}$$

Where, y represents the predicted AQI value, b_0 is the intercept, $b_1, b_2, \ldots, b_i$ are the regression coefficients, and $x_1, x_2, \ldots, x_i$ represent pollutant concentrations such as PM2.5, NO_2, and CO. This equation shows how each pollutant contributes to the overall AQI prediction.

B. Random Forest

This model is an ensemble learning technique that improves prediction performance by combining multiple decision trees. Each tree is trained on a randomly selected subset of the data, and the final prediction is obtained by averaging the outputs of all trees. This aggregation process reduces variance and enhances generalization ability. In AQI prediction, Random Forest is effective in handling complex, non-linear relationships among pollutant variables.

$$y = \frac{1}{K}\sum_{k=1}^{K} h_k(x) \tag{2}$$

Where, y represents the predicted output value, K is the total number of weak learners, and $h_k(x)$ is the prediction of the kth base learner for input x. This equation represents averaging the outputs of multiple learners to improve prediction accuracy.

C. XGBoost

It is an advanced gradient boosting algorithm that constructs models in a sequential manner. Each function ft(x) represents a decision tree that is added to correct the residual errors of the previous trees. The final prediction is obtained by summing the outputs of all trees. This iterative error-correction mechanism allows XGBoost to achieve high accuracy while controlling overfitting through regularization techniques. It is widely used for structured data problems such as AQI prediction.

$$y = \sum_{t=1}^{T} f_t(x) \tag{3}$$

Where, y represents the predicted output value, T is the total number of boosting stages, and $f_t(x)$ is the prediction of the tth learner. This equation shows how multiple weak learners are combined sequentially to form a strong predictor.

D. CatBoost

This is a gradient boosting algorithm designed to efficiently process datasets containing categorical features. It introduces an ordered boosting technique, where a permutation (pi) of the dataset is used to reduce prediction bias and prevent data leakage. Each function ft(x,pi) represents a tree built during the boosting process. The model also uses symmetric trees, which enhance computational efficiency. In AQI prediction, CatBoost performs well due to its ability to handle categorical variables such as city information without extensive preprocessing.

$$y = \sum_{t=1}^{T} f_t(x, \pi) \tag{4}$$

Where, y represents the predicted output value, T is the total number of trees or learners, $f_t(x, \pi)$ is the prediction of the tth learner with parameter set π, and x denotes the input data. This equation demonstrates ensemble learning with parameterized models.

E. AdaBoost

AdaBoost, or Adaptive Boosting, is an ensemble learning method that combines multiple weak learners to form a strong predictive model. Each weak learner ht(x), typically a shallow decision tree, is assigned a weight based on its performance. During training, the algorithm focuses more on incorrectly predicted samples by increasing their importance in subsequent iterations. The final prediction is computed as a weighted sum of all learners. This approach helps improve model accuracy by iteratively reducing errors.

$$y = \sum_{t=1}^{T} \alpha_t h_t(x) \tag{5}$$

Where, y represents the predicted output value, T is the total number of weak learners, α_t is the weight assigned to the tth learner, and $h_t(x)$ is the prediction of the tth learner for input x. This equation indicates that stronger learners receive higher weights in the final prediction.

F. SVM(Support Vector Machine)

SVM is a supervised learning algorithm that aims to find an optimal boundary for prediction tasks. The model represents the prediction using a function defined by a weight vector w and bias b. it attempts to maximize the margin between data points while minimizing prediction error. SVM can handle both linear and no-linear relationships through the use of kernel functions, making it suitable for complex AQI datasets.

$$f(x) = w^T x + b \tag{6}$$

Where, $f(x)$ represents the decision function, w is the weight vector, $w^T x$ is the dot product of weights and input features, x is the feature vector, and b is the bias term. This equation defines a linear decision boundary used for classification or regression tasks.

V. METHODOLOGY

A. Dataset Description

The dataset we have used is obtained from Kaggle and contained cleaned air quality data collected from multiple cities across India. It provides detailed information about various air pollutants along with corresponding Air Quality Index(AQI) values. The dataset is structured and preprocessed, making it suitable for machine learning applications.

The data consists of several important pollutant parameters, including particulate matter (PM2.5 and PM10), nitrogen oxides (NO, NO2, and Nox), ammonia(NH3), carbon monoxide(CO), sulfur dioxide(SO2), and ozone(O3). These pollutants are widely recognized as key indicators of air quality and play a significant role in determining AQI levels. Each record represents air quality measurements for a specific city at a given date and time.

In addition to pollutant concentrations, the dataset includes categorical information such as the city name and temporal features such as date and time. These attributes help capture spatial and temporal variations in air quality patterns. The dataset also provides the AQI values, which is used as the target variable for regression tasks, and an AQI category label, which can be used for classification purposes.

The dataset has been preprocessed to handle missing values and inconsistencies, ensuring better model performance and reliability. Since it is already cleaned, minimal preprocessing is required before applying machine learning algorithms. This makes the dataset well-suited for building and evaluating predictive models for AQI estimation.

B. Data Acquisition

For this project, I used an air quality dataset which I have taken from Kaggle [7]. The name of the dataset is Cleaned Air Quality India. It contains pollution data of different cities in India. I chose this dataset because it is relevant to my project and already arranged in a usable format.

The dataset was downloaded as a CSV file. After downloading, it was uploaded to Google Colab where the complete coding work was done. Python language was used for reading and processing the file. Libraries such as Pandas and NumPy were used during this step.

The file contained values of important pollutants like PM2.5, PM10, NO2, SO2, CO and O3. It also had AQI values In this research work, pollutant values were taken as input features and AQI was used as the output variable for prediction.

As the dataset was already cleaned, only small checks were needed before model training. Missing values were verified, column names were reviewed, and required features were selected. Extra columns which were not useful for prediction were ignored.

After this, the dataset was divided into training data and testing data. The training portion was used to build machine learning models, while the testing portion was used to check how well the models performed on unseen data.

Using a public dataset is useful because the same work can be repeated by other researchers. It also makes comparison easier for future studies.

TABLE I
XGBOOST HYPERPARAMETERS

Parameter	Value	Description
n_estimators	500	Number of boosting trees
learning_rate	0.03	Step size for learning
max_depth	7	Maximum depth of each tree
subsample	0.8	Fraction of training samples used
colsample_bytree	0.8	Fraction of features used per tree

VI. MATHEMATICAL ANALYSIS OF XGBOOST

XGBoost is a supervised machine learning technique widely used for regression and classification problems. It is based on the boosting concept, where several weak decision trees are combined to create a stronger prediction model. Instead of building all trees at once, the trees are created one after another. Each new tree learns from the errors made by the previous trees and tries to improve the overall result.

For AQI prediction, the model starts with an initial estimate. After that, it compares the predicted AQI with the actual AQI values present in the dataset. The difference between these values is treated as error. A new tree is then trained to reduce this error. This process continues for multiple rounds until the model reaches better accuracy.

An important advantage of XGBoost is regularization. It controls the complexity of trees and prevents the model from becoming too dependent on training data. Because of this, the chances of overfitting are reduced. The model also includes a learning rate, which controls the contribution of every new tree added during training. Lower learning rates generally give more stable performance.

XGBoost is considered effective for AQI datasets because air pollution depends on many pollutants such as PM2.5, PM10, NO2, CO, and SO2. These variables often have nonlinear relationships. XGBoost can capture such patterns efficiently and provide reliable predictions.

Due to its speed, accuracy, and ability to handle structured data, XGBoost is commonly used in environmental forecasting and AQI estimation studies.

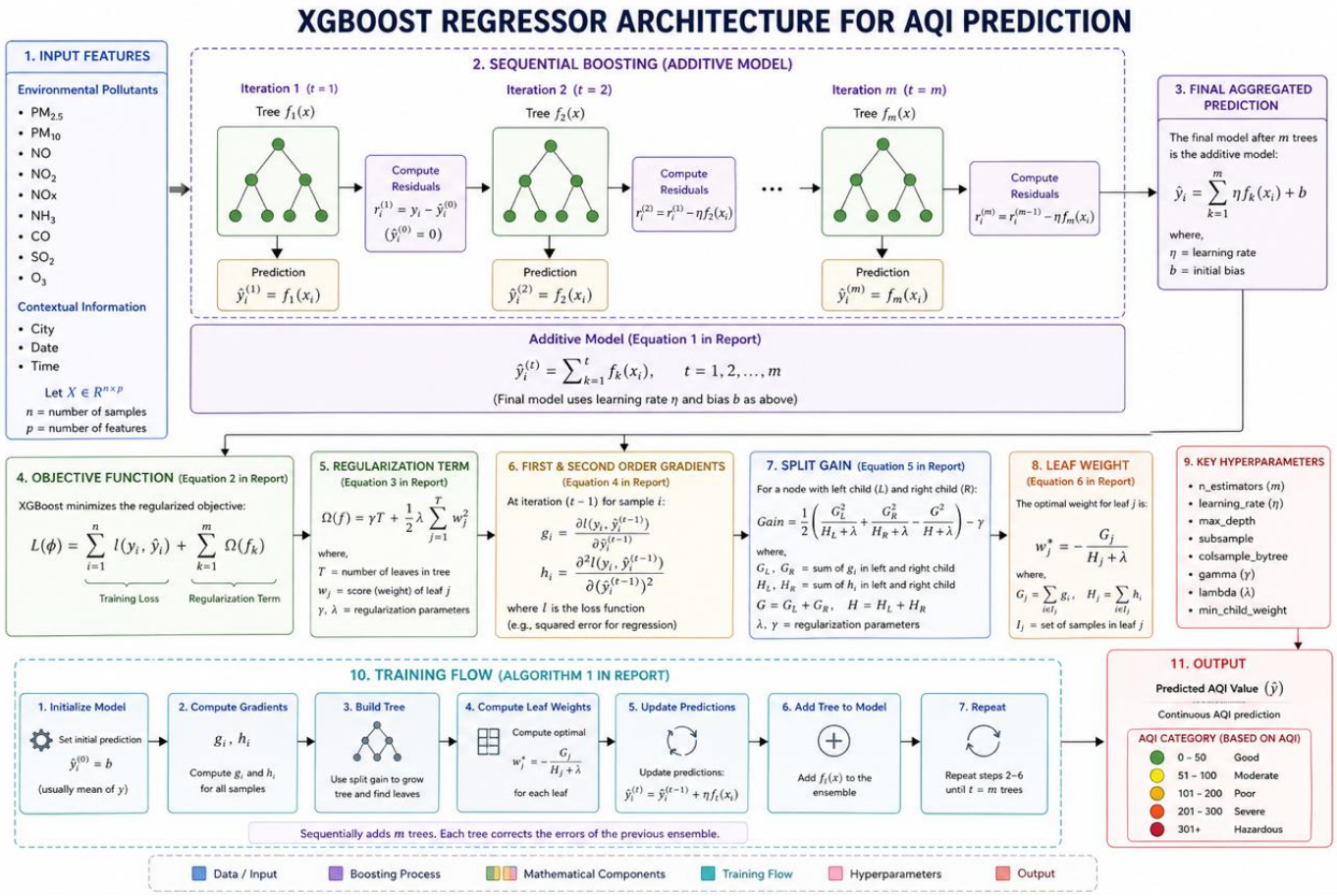


Fig. 2. XGBoost Architecture

VII. XGBoost Algorithm

B1. Mathematical Formulation of XGBoost

The Extreme Gradient Boosting (XGBoost) model is an ensemble learning method based on additive tree boosting. It combines multiple regression trees in a sequential manner, where each new tree is trained to reduce the prediction error of the previous trees. The predicted output at iteration t is given by:

$$\hat{y}_i^{(t)} = \sum_{k=1}^{t} f_k(x_i), \quad f_k \in \mathcal{F} \tag{7}$$

where $\mathcal{F}$ represents the space of regression trees, f_k denotes an individual tree, and t is the boosting round.

Objective Function

The training objective of XGBoost is to minimize a regularized loss function:

$$\mathcal{L}(\phi) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{8}$$

where:

- $l(y_i, \hat{y}_i)$ is the loss between actual and predicted AQI values.
- $\Omega(f_k)$ is the regularization term used to control model complexity.
- K is the total number of trees.

Regularization Term

To avoid overfitting, XGBoost penalizes overly complex trees using:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{9}$$

where:

- T is the number of leaves in the tree,
- w_j is the score assigned to leaf j,
- γ controls the penalty for adding leaves,
- λ is the L2 regularization parameter.

Second-Order Approximation

For faster optimization, XGBoost uses second-order Taylor expansion:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{n} \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t) \tag{10}$$

where:

$$g_i = \frac{\partial l(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}} \tag{11}$$

$$h_i = \frac{\partial^2 l(y_i, \hat{y}_i^{(t-1)})}{\partial (\hat{y}_i^{(t-1)})^2} \tag{12}$$

Here, g_i and h_i are the first and second order derivatives of the loss function.

Algorithm 1: XGBoost for AQI Prediction

Input: Dataset $D = \{(x_i, y_i)\}$, boosting rounds K, learning rate η, regularization parameters γ, λ

Output: Final prediction model $\hat{y}_i$

Initialize predictions:

$$\hat{y}_i^{(0)} = 0 \quad \forall i$$

for $t = 1$ to K **do**

Compute first-order gradients:

$$g_i = \frac{\partial l(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}}$$

Compute second-order gradients:

$$h_i = \frac{\partial^2 l(y_i, \hat{y}_i^{(t-1)})}{\partial (\hat{y}_i^{(t-1)})^2}$$

Construct a regression tree $f_t(x)$

Calculate split gain:

$$Gain = \frac{1}{2}\left(\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda} \right) - \gamma$$

Select split with maximum gain

Compute leaf weight:

$$w_j = -\frac{G_j}{H_j + \lambda}$$

Update prediction:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta f_t(x_i)$$

end for

Final prediction:

$$\hat{y}_i = \sum_{t=1}^{K} \eta f_t(x_i)$$

Description

XGBoost improves prediction accuracy by building trees sequentially, where each tree learns from previous errors. It uses regularization to reduce overfitting and second-order derivatives for faster optimization. Because AQI data often contains nonlinear relationships among pollutants, XGBoost is highly suitable for air quality prediction tasks.

VIII. Performance Evaluation Metrices

Machine learning models used for AQI prediction were assessed using several statistical performance measures. These metrics help compare prediction accuracy, error level, and computational efficiency of different algorithms. In this study, Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R2), Mean Absolute Percentage Error (MAPE), and execution Time were selected for evaluation.

A. Root Mean Square Error (RMSE)

RMSE measures the average magnitude of prediction error by taking the square root of squared differences between actual and predicted AQI values. It gives higher importance to large errors, making it useful when major deviations must be penalized. A lower RMSE value indicates better prediction performance.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{13}$$

B. Mean Absolute Error (MAE)

MAE calculates the average absolute difference between observed and predicted AQI values. It provides a clear interpretation of average error in the same unit as the target variable. Lower MAE values represent more accurate model predictions.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{14}$$

C. Coefficient of Determination (R 2)

The R 2 score indicates how well the model explains the variation in AQI values. Its value generally lies between 0 and 1, where values closer to 1 represent stronger predictive capability. A higher R 2 value suggests that the model fits the dataset effectively.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{15}$$

D. Mean Absolute Percentage Error (MAPE)

MAPE expresses prediction error as a percentage. It helps compare model accuracy on a relative scale and is easy to understand. Smaller MAPE values indicate better forecasting performance.

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{16}$$

E. Computational Time

Execution time represents the total duration required to train and evaluate the model. It is important when selecting models for real-time AQI monitoring systems. Lower computational time reflects faster model performance.

$$T = t_{end} - t_{start} \tag{17}$$

IX. TRAINING

The dataset used in this study includes air quality parameters collected from different monitoring stations. It contains important pollutants like PM2.5, PM10, NO, SO, and CO, which directly affect the Air Quality Index (AQI).

Before training the models, the data was cleaned by handling missing values and removing any inconsistent entries. Only relevant numerical features were used for building the models. After preprocessing, the dataset was divided into training and testing sets in an 80:20 ratio.

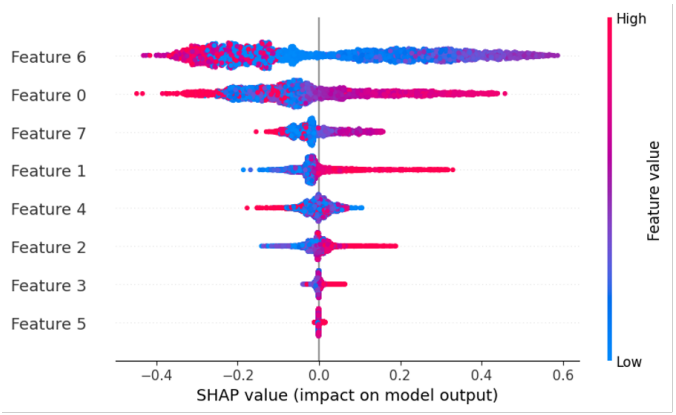


Fig. 3. SHAP Value

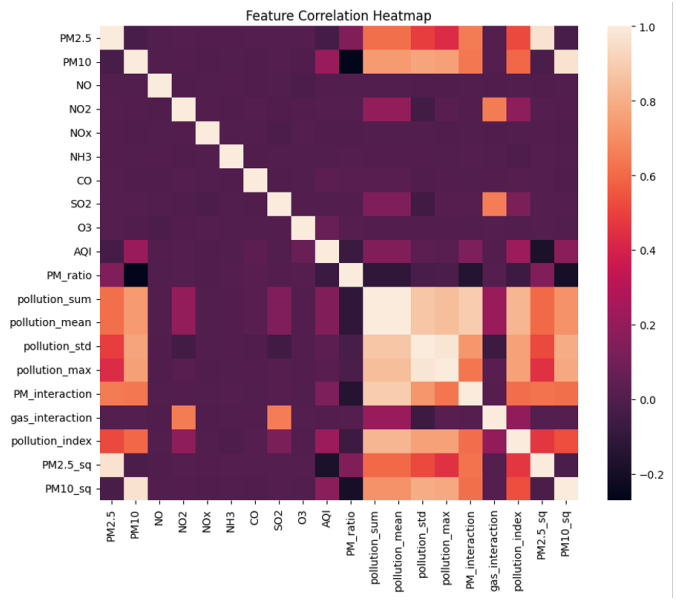


Fig. 4. Feature Correlation Heatmap

Different machine learning models were then trained, including Linear Regression, SVM, AdaBoost, XGBoost, and CatBoost. For Linear Regression and SVM, feature selection and scaling were applied to improve performance. Polynomial features were also added to Linear Regression to capture non-linear patterns.

On the other hand, tree-based models like XGBoost and CatBoost were trained directly on the dataset without scaling, as they can handle complex relationships on their own. AdaBoost was trained using a decision tree as its base model.

Overall, this approach helped in comparing different models and identifying the most effective one for AQI prediction.

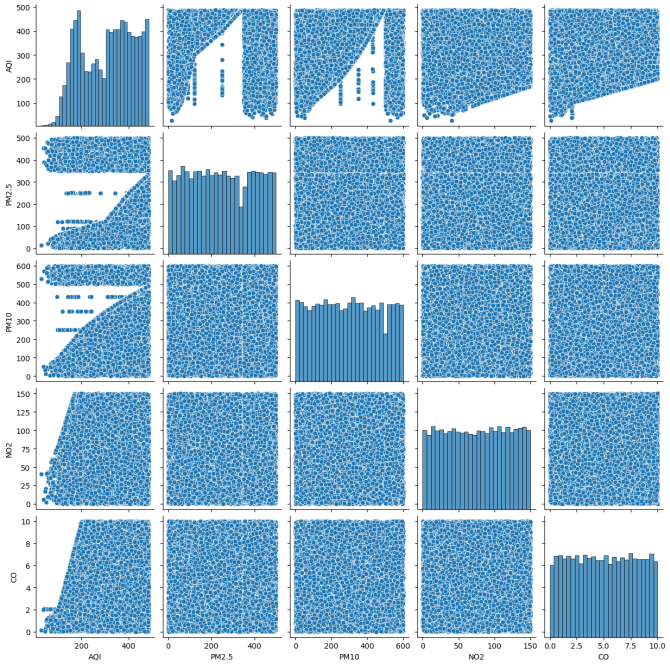


Fig. 5. Pair Plot

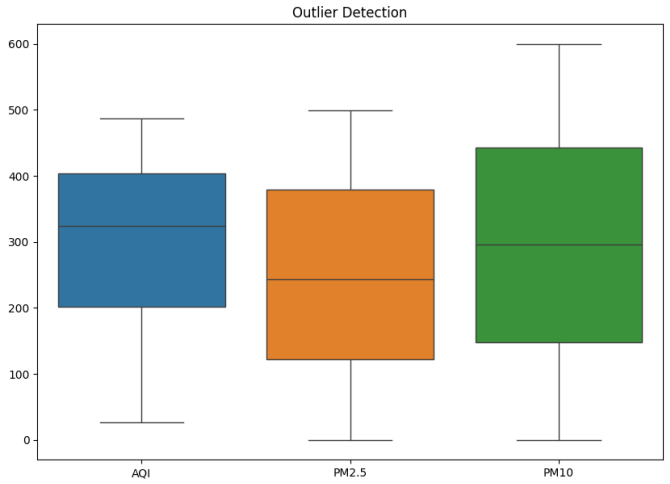


Fig. 6. Outlier Detection

X. CONTRIBUTION

Several significant contributions have been made in the context of air quality prediction using machine learning in this study. These are discussed below:

To begin with, the paper involves a comparison between multiple machine learning models employed in predicting AQI. The models used include Linear Regression, , Support Vector Machine, XGBoost, and CatBoost. Different classes of machine learning models – namely traditional machine learning models and ensemble learning models – have been used to compare their effectiveness.

Furthermore, a full-fledged machine learning pipeline has been established in the paper. The stages of this pipeline include data preprocessing, model training, and finally the evaluation of the results obtained from the models. Data preprocessing entails dealing with missing data, inconsistency, scaling the features, and separating training data from testing data.

An important contribution made by the paper is the design and testing of XGBoost and CatBoost, which are two advanced boosting algorithms that can capture non-linear and complex dependencies between air pollutants, including PM2.5, PM10, NO2, and CO.

Also, the researchers show that advanced boosting algorithms are more accurate in predicting AQI values than traditional methods, making them more suitable for real-world applications. They find that the XGBoost model is the most accurate with an R2 score of approximately 0.9775 and the smallest RMSE of approximately 16.55.

Moreover, the paper includes an analysis and discussion of the accuracy and efficiency of the XGBoost and CatBoost models using appropriate metrics. Such information will help in comparing the performance of various models and determining which one is better for solving the problem.

Furthermore, the research highlights the shortcomings of traditional models, such as linear regression and SVM, in processing multidimensional data and identifying non-linear dependencies between variables. This information proves the need to transition from traditional methods to modern ML tools.

Lastly, this paper has a practical impact on the field since the results can be applied in environmental monitoring systems to predict AQI values, issue warnings to citizens, and implement pollution control measures.

XI. RESULTS AND DISCUSSION

This study looked at how four machine learning models. Linear Regression, Support Vector Machine, XGBoost and CatBoost -are predicting the Air Quality Index. We used two measures to see how good these models are: Root Mean Square Error and the coefficient of determination. These measures help us understand how accurate the Air Quality Index models are and if we can really trust them. The Air Quality Index models were then checked to see what the results really mean. How well each of the four models. Linear Regression, Support Vector Machine, XGBoost and CatBoost. Is working.

A. Performance Evaluation

The results of the experiments show that there is a difference in how well the models work. XGBoost works the best with the error rate of 16.55 and the highest score of 0.9775. This means XGBoost can explain about 97.75 percent of the difference in Air Quality Index values. In this when we compared XGBoost has the score and the lowest error rate . CatBoost also works well with an error rate of 17.60 and a score of 0.9745. This shows that models that use boosting are very good at

predicting Air Quality Index values. Catboost does not work as well as XGBoost. On the hand Support Vector Machine works okay with an error rate of 53.23 and a score of 0.7674. This means it has a time understanding the complicated relationships in Air Quality Index data. Linear Regression works the worst with an error rate of 61.97 and a score of 0.6847. This shows that it cannot find the patterns that're not straight lines in environmental data. XGBoost and CatBoost are the models for predicting Air Quality Index values because they can understand the complicated relationships, in the data.

B. Prediction Quality Analysis

The superior performance of XGBoost can be attributed to several factors:

1) Actual vs Predicted • The points on the graph are very close to the line • This shows that XGBoost predictions are accurate
2) We also looked at the Residual Plot. Saw that the mistakes are spread out randomly around zero without any pattern. This means that the model does not have any bias and works well with data.

C. Justification for Best Model Selection

The strong performance of XGBoost can be explained through the following key factors:

1) Ability to Capture Non-Linear Relationships AQI depends on multiple pollutants such as PM2.5, PM10, NO, and CO, whose relationships are highly complex and non-linear. XGBoost, being a tree-based model, is able to effectively capture these patterns, leading to more accurate predictions compared to traditional linear models.
2) Automatic Learning of Feature Interactions In environmental data, pollutants do not act independently; their combined effects influence AQI. XGBoost naturally learns these interactions during training, without requiring manual feature engineering, which improves overall model performance.
3) Regularization for Better Generalization XGBoost includes built-in regularization techniques that help control overfitting. This ensures that the model performs consistently not only on training data but also on unseen data, which is important for real-world AQI prediction.
4) Robustness to Data Imperfections Air quality datasets often contain missing values, noise, and outliers. XGBoost handles these issues efficiently, reducing the need for heavy preprocessing while still maintaining strong performance.

Overall, these characteristics make XGBoost a reliable and practical model for AQI prediction, especially in real-time monitoring systems and smart city applications where accuracy and robustness are essential.

D. Feature Importance and SHAP Analysis

The interpretability of the model was analyzed using feature importance and SHAP (SHapley Additive exPlanations), which provide insights into how different input features influence AQI prediction.

1) Feature Importance Analysis The feature importance plot (Fig. 5) highlights the most influential variables contributing to AQI prediction. It is observed that features related to particulate matter, such as $PM2.5^2$, $PM10^2$, and PM interaction terms, have the highest importance scores. This indicates that particulate pollutants play a dominant role in determining AQI levels. The presence of squared and interaction features among the top contributors also suggests that the relationship between pollutants and AQI is highly non-linear and influenced by combined effects rather than individual factors alone.
2) SHAP Summary Analysis The SHAP summary plot (Fig. 3) provides a global interpretation of feature impact on model predictions. It shows that features like PM2.5 and PM10 consistently have high SHAP values, indicating their strong influence on AQI. In the plot, higher feature values (represented in red) are associated with positive SHAP values, meaning they contribute to an increase in predicted AQI. This demonstrates a direct relationship between pollutant concentration and air quality degradation. Overall, the SHAP results confirm that as pollution levels increase, AQI also increases, which aligns with established environmental understanding. This consistency indicates that the model's predictions are not only accurate but also scientifically meaningful.
3) SHAP Waterfall Analysis The SHAP waterfall plot provides a local explanation of individual predictions by showing how each feature contributes to the final AQI value for a specific instance. This level of interpretability helps in understanding the decision-making process of the model for single predictions. It clearly illustrates how different pollutants either increase or decrease the predicted AQI.

Such explanations enhance the transparency of the model and demonstrate that it is not a complete "black box." Instead, its predictions can be interpreted and trusted, which is especially important in environmental applications where decision-making impacts public health.

E. Comparative Analysis of Other Models

To better understand the effectiveness of the proposed approach, the performance of XGBoost was compared with other models.

1) CatBoost achieved an RMSE of 17.60, which is very close to XGBoost (16.55), indicating that boosting-based methods perform well for AQI prediction. However, its slightly lower performance may be due to the dataset being mostly numerical, whereas CatBoost is more optimized for categorical data.
2) SVM showed moderate performance with an RMSE of 53.23, which is significantly higher than boosting models. This suggests that it struggles to capture the complex and high-dimensional relationships present in

TABLE II
PERFORMANCE COMPARISON OF MODELS

Model	Dataset	R^2	RMSE	MAPE (%)	MAE	Time (s)
Linear Regression	Training	0.717773	58.581594	15.353751	41.679236	0.179805
	All Dataset	0.719814	58.382887	15.336193	41.647803	0.179805
	Testing	0.727941	57.581198	15.265965	41.522072	0.179805
SVM	Training	0.778452	51.903319	13.973595	39.888892	52.324372
	All Dataset	0.778471	51.913219	13.967111	39.950789	52.324372
	Testing	0.778528	51.952798	13.941177	40.198375	52.324372
CatBoost	Training	0.976825	16.786800	4.323279	9.100977	2.745643
	All Dataset	0.976376	16.952836	4.369646	9.191308	2.700000
	Testing	0.974579	17.601324	4.555115	9.552634	2.745643
XGBoost	Training	0.992069	9.820406	2.404489	4.996149	1.235078
	All Dataset	0.989151	11.488321	2.729888	5.528271	1.235078
	Testing	0.977505	16.557357	4.031484	7.656758	1.235078

AQI data. Additionally, its sensitivity to parameters such as C and gamma makes it less stable and harder to optimize.

3) Linear Regression performed the worst, with an RMSE of 61.97, indicating poor prediction accuracy. This is mainly because it assumes a linear relationship between features and AQI, whereas the actual relationships are non-linear, leading to underfitting.

Overall, the comparison clearly shows that boosting models outperform traditional approaches, with XGBoost providing the best balance between accuracy and reliability.

F. Data Behavior Analysis

The underlying characteristics of the dataset further support the observed model performance. The correlation heatmap (Fig. 4) shows strong relationships among different pollutants, indicating that AQI is influenced by multiple interdependent variables rather than isolated factors. This interdependence highlights the need for models capable of handling complex, non-linear relationships. Similarly, the pairplot (Fig. 5) reveals non-linear and scattered relationships between key features such as PM2.5, PM10, and AQI. The absence of clear linear patterns explains why Linear Regression performs poorly, as it is unable to capture such complexity in the data.

G. Challenges and Limitations

Despite achieving strong results, certain limitations were identified in this study. AQI prediction is inherently complex due to the non-linear interactions among multiple pollutants. Models like SVM require careful hyperparameter tuning, which can be computationally intensive and time-consuming. Although boosting models such as XGBoost and CatBoost perform well, they may still be prone to overfitting if not properly regularized. Additionally, the current approach does not consider temporal dependencies in AQI data, which could further improve prediction accuracy if incorporated using time-series or deep learning models.

H. Conclusion of Discussion

Based on both numerical results and graphical analysis, XGBoost emerges as the most accurate and reliable model for AQI prediction. It achieves the lowest RMSE of 16.55 and the highest R^2 score of 0.9775, indicating excellent predictive performance. The graphical results further support these findings by demonstrating high prediction accuracy, low error distribution, and strong generalization capability. Overall, the study clearly shows that ensemble boosting methods outperform traditional approaches in modeling AQI data. In conclusion, this study demonstrates that advanced ensemble techniques, particularly XGBoost, are highly effective for AQI prediction due to their ability to capture complex and non-linear relationships present in environmental data.

XII. CONCLUSION

This paper aims at evaluating different machine learning methods for estimating Air Quality Index (AQI) using air pollution datasets. Algorithms implemented in this evaluation process were Linear Regression, Support Vector Machine (SVM), XGBoost, and CatBoost. In this research, the main aim was choosing the best model among these methods based on an understanding of the relationship between different pollutants to forecast AQI. RMSE and R^2 score metrics were used for efficiency evaluation of the models.

Based on the experimental results, it is evident that there is a significant difference between the accuracy of the models. Among all these models, the superiority of XGBoost is obvious since the algorithm had the minimum value of RMSE (16.55) and the maximum R^2 score (0.9775). The model indicates that nearly 97.75 percent of the variance in AQI is explained by the model. The CatBoost also performed very effectively in the prediction task and proved that it is possible to predict AQI effectively using boosting-based algorithms. Although SVM also performed quite effectively, it failed to work in large-scale complex situations. On the other hand, linear regression also did not provide satisfactory results

since the algorithm assumes linearity while AQI is based on nonlinearity.

Such an advantage of XGBoost is linked to its ability to model more complex non-linear correlations between such pollutants as PM2.5, PM10, NO, SO, and CO. Besides, the model can automatically learn feature interaction and utilize regularization to prevent overfitting and increase accuracy. In summary, this paper proved that ensemble and boosting-based algorithms are much more efficient compared to classic approaches for the purpose of AQI prediction. Additionally, proper data pre-processing and RMSE/R^2 evaluation is critical for obtaining valid predictions. The presented models are applicable in such real-life applications as air pollution monitoring and health warning systems.

At the same time, there are several shortcomings associated with the proposed research. For instance, model's generalization abilities can be influenced by the quality and quantity of the used dataset and certain external factors such as location and seasonality. Furthermore, the study ignores possible time-related dependencies between AQI observations. Potential areas of further research include the application of real-time Internet-of-Things and satellite data, spatiotemporal features, and models of deep learning, for example, LSTM and CNN models, as well as SHAP/LIME techniques.

To summarize, it can be stated that this study has established that the application of modern machine learning methods such as boosting is very efficient for AQI prediction tasks. The proposed algorithms demonstrate high levels of precision, scalability, and robustness.

XIII. FUTURE SCOPE

Future improvements on the AQI prediction algorithm will involve working with bigger and varied sets of data coming from different areas. Live data from IoT sensors and satellites will be incorporated in the system to provide real-time monitoring and alerts about the AQI levels. More sophisticated deep learning algorithms like Long Short-Term Memory, Convolutional Neural Networks, and combination algorithms will enhance the predictions. Spatiotemporal features can be incorporated into the model to account for the variations based on location and time. Explainable AI methods like SHAP and LIME can be applied as well.

REFERENCES

[1] Hajat, A., Hsia, C., & O'Neill, M. S. (2015). Socioeconomic disparities and air pollution exposure: A global review. *Current Environmental Health Reports, 2*(4), 440–450.

[2] Kelly, F. J., & Fussell, J. C. (2015). Air pollution and public health: Emerging hazards and improved understanding of risk. *Environmental Geochemistry and Health, 37*(4), 631–649.

[3] Li, X., Peng, L., Hu, Y., Shao, J., & Chi, T. (2017). Long short-term memory neural network for air pollutant concentration predictions. *Atmospheric Environment, 158*, 166–175.

[4] de Hoogh, K., et al. (2018). Modelling daily PM2.5 concentrations at high spatio-temporal resolution using satellite data and land-use variables. *Environmental Pollution*.

[5] Zhao, C., et al. (2020). Estimating daily PM2.5 concentration using Random Forest model. *Environment International*.

[6] Samad, A., et al. (2023). Air pollution prediction using machine learning techniques: A review. *Atmospheric Environment*.

[7] Gryech, I., et al. (2024). Applications of machine learning and IoT for air quality prediction: A review. *Engineering Applications of Artificial Intelligence*.

[8] Bai, X., et al. (2024). Prediction of PM2.5 concentration based on CNN-LSTM model. *PeerJ Computer Science*.

[9] Singh, R., et al. (2021). Support Vector Machine based urban air quality prediction. *Environmental Monitoring and Assessment*.

[10] Kumar, P., & Goyal, R. (2020). Artificial Neural Network approach for air pollutant forecasting. *Journal of Cleaner Production*.

[11] Chen, Y., et al. (2022). XGBoost framework for accurate air quality prediction. *Atmospheric Research*.

[12] Wang, L., et al. (2021). Comparative study of ensemble learning methods for AQI prediction. *Environmental Science and Pollution Research*.

[13] Sharma, V., et al. (2023). Air quality prediction in Indian metropolitan cities using machine learning. *Sustainable Cities and Society*.

[14] Patel, M., et al. (2024). IoT sensor integrated machine learning model for real-time AQI monitoring. *Smart Cities*.

[15] Kaggle (Dataset):- https://www.kaggle.com/datasets/sathvikisikell